Jazz Graph:
A Graph Neural Network for Jazz Music Recommendation

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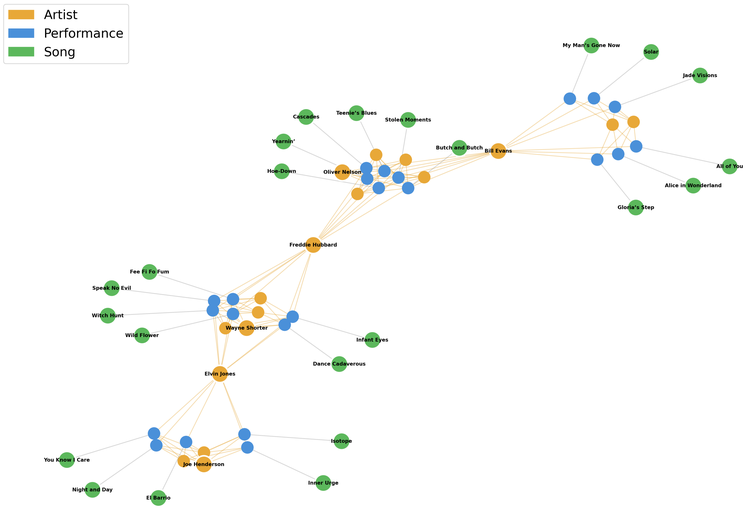
Problem Statement

Create musical recommendation by encoding collaboration as similarity signal.

Jazz Data as Graph



A sub-graph illustrating connections among jazz performances. This is the one-hop neighborhood of the depicted performances.

Data

Node/Edge Type	Count	Mean degree source	Mean degree destination
Artist	24,470		
Performance	130,695		
Song	31,828		
Artist *performs* performance	516,713	21.11	3.95
Performance *performing* song	86,588	.66	2.72
Artist *composed* song	36,252	1.48	1.13

Graph Neural Network

JazzGraph GNN is a three-layer network for a heterogeneous graph; it combines GraphSAGE for featureless edges and GATv2Conv for artist-performance edges, which have an instrument feature.

Training Task

Construct batches of positives pairs, where x_i, x_j are positive if they are songs on the same album. Our loss function is computed as normalized temperature scaled cross-entropy loss:

$$\mathcal{L}_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2n} [k \neq i] \exp(\text{sim}(z_i, z_j)/\tau)}$$

Where sim is cosine similarity and $[k \neq i]$ is an indicator function equal to 1 unless $k = i$, when it is 0.

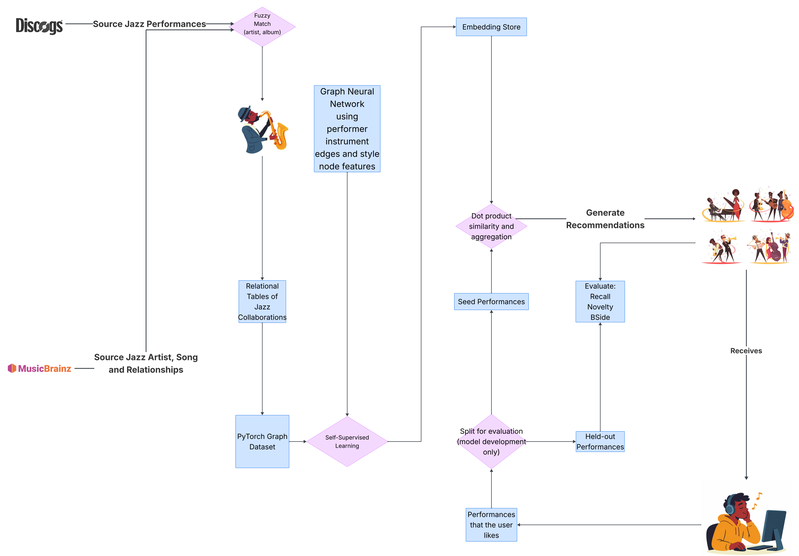
Recommendation

Score each member of the performance corpus, p_i, with each seed performance (user-liked recording)

$$score(p_i) = agg(\sum_{k=1}^{n} gnn(p_i)^{\intercal} gnn(p_k))$$

where *gnn* is the embedding learned from the graph of jazz performances, artists and songs and *agg* is an aggregation or pooling function. Experiments were run with *max*, *sum* and *softmax-weighting*.

System Architecture



A data extraction pipeline(left), self-supervised learning with GNN (middle) and recommendation system (right.)

Repository

https://www.github.com/philosofool/jazz-graph

Results

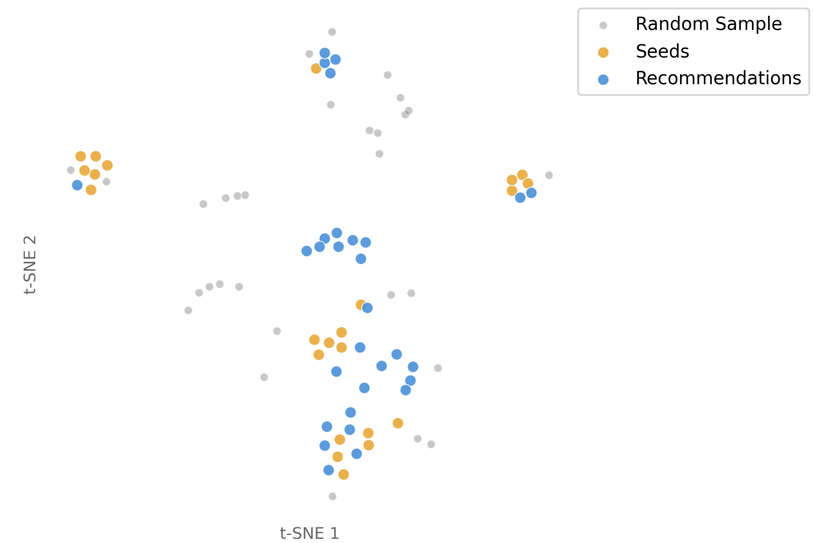
- Novel Recall: Relevance of model recommendations.
- Novelty: Exploratory model behavior.
- B-Side: Expected recommendations for narrow seeds.

Task	Features	Pooling	Novel Recall	Familiar Recall	Novelty	B-side Mean
Combined Loss	Instrument, Style	sum	0.276	0.328	0.806	0.822
Combined Loss	Instrument, Style	max	0.297	1	0.840	**0.839**
Combined Loss	Instrument, Style	softmax	0.276	0.318	0.807	0.822
Edge Ablation	Instrument, Style	sum	0.314	0.365	0.910	0
Edge Ablation	Instrument, Style	max	0.239	1	**0.917**	0
Edge Ablation	Instrument, Style	softmax	0.323	0.393	0.910	0
Match Album	None	sum	0.278	0.378	0.850	0.434
Match Album	None	max	0.173	1	0.835	0.345
Match Album	None	softmax	0.314	0.395	0.840	0.411
Match Album	Instrument, Style	sum	0.485	0.544	0.805	0.548
Match Album	Instrument, Style	max	0.295	1	0.829	0.523
Match Album	Instrument, Style	softmax	**0.518**	**0.608**	0.804	0.548
Random Walk Baseline	—	—	0.194	0.202	0.523	0.132
Artist Weighted Baseline	—	—	0.420	0.405	0.011	0

Conclusions

- Collaboration encodes useful similarity information.
- Recommendation strength depends on aggregation.
- Diversity in learned system is much better than baselines.
- Many opportunities to improve the system.

Jazz Performance Embedding Space



t-SNE plot of the recommendations and seeds performances using 4 prominent jazz albums.

Future Work

- Expand validations
- Provide additional features, such as session dates, mood, tempo
- Encode recordings as features
- Learn with Metapath2Vec or another algorithm
- Deploy system for public use

Discovery

Listen to recommendations based on Joe Henderson's "Isotope":

Spotify

Apple Music

Lester Bowie, "The Great Pretender" (1982) Great music I discovered with JazzGraph.